Adding and Subtracting in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 1 — Adding and Subtracting in Scientific Notation	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Add and subtract numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. The two exponents stay within 2 of each other — that is this benchmark's own limit. Multiplication and division are 4.02–03; significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the powers of ten are set against the place-value names, so 'the powers must match' is read off the structure rather than announced as a rule.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 4 — the coefficient is checked against $1 \le a < 10$ before the board goes up, in both directions.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 1.23×10^7 and 12.3×10^6, which have the same value, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must two numbers written in scientific notation carry the same power of ten before they can be added?

Building on. MA.8.NSO.1.4 (3.08–3.09) — scientific notation, the coefficient rule and reading notation back; Grade 6 — adding and subtracting decimals by place value.

Working toward. 4.02–03 — multiplying and dividing in scientific notation; MA.8.NSO.1.6 (4.04) — real-world problems and significant digits.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why can 5.2×10^6 and 3.1×10^6 be added straight away, while 5.2×10^6 and 3.1×10^5 cannot?	DOK 2
Devon adds the exponents and gets 10^6. What would have to be true about the expression for adding exponents to be the right move?	DOK 3
12.3×10^6 and 1.23×10^7 are the same number. Why does one of them count as an answer and the other does not?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3.2 \times 10^5) + (4.1 \times 10^5)$	***7.3 × 10⁵***	7.3×10^{10} — the exponents were added as well as the coefficients [B1G-M 8.NSO.1.5 misconception: exponents added]; 1.312×10^{11} — the numbers were multiplied [8.NSO.1.5: the operation is addition]
2	A stadium sold 4.2×10^4 tickets on Friday and 3.5×10^4 on Saturday. Each ticket cost 38 dollars. How many tickets were sold on the two days?	***7.7 × 10⁴***	anything built from 38 — the unneeded figure was used [the question asks for a count, not a cost]; 7.7×10^8 — the exponents were added [B1G-M 8.NSO.1.5 misconception]
3	$(8 \times 10^{-6}) + (3 \times 10^{-6})$	***1.1 × 10⁻⁵***	11×10^{-6} — right value, but 11 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; 1.1×10^{-11} — the exponents were added, giving 11×10^{-12}, and the answer was then finished
4	$(5.6 \times 10^8) - (4.9 \times 10^8)$	***7 × 10⁷***	0.7×10^8 — right value, but 0.7 is not at least 1 [8.NSO.1.5: the coefficient satisfies $1 \le a < 10$]; 1.05×10^9 — added instead of subtracting
5	$(1.6 \times 10^4) + (2.3 \times 10^6)$. Which is the sum?	***C — 2.316 × 10⁶***	(A) the coefficients were added without matching the powers of ten first [8.NSO.1.5: the powers must match]; (B) the coefficients were added and the exponents were added [B1G-M 8.NSO.1.5 named misconception: $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$]; (D) 1.6×10^4 was rewritten as 0.16×10^6 — the decimal point moved one place too few [8.NSO.1.5: the point moves as many places as the exponent changes]

#	What the board asks	Answer	If the board is wrong
6	$(2.5 \times 10^6) + (9.8 \times 10^6)$. Which is the sum, written in scientific notation?	***D — 1.23 × 10⁷***	(A) equal in value, but 12.3 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (B) the exponents were added as well as the coefficients [8.NSO.1.5: the power of ten does not change]; (C) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]
7	$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$	***7.16 × 10⁻²***	1.4×10^{-2} — the coefficients were subtracted without matching the powers [8.NSO.1.5]; 7.16×10^{-5} — the exponents were added, −2 + (−3) [8.NSO.1.5: the power of ten is a label, not a term]
8	$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$. What is the value of n?	***n = 5.7***	−3.48 — the target was not rewritten over 10^6 before subtracting [8.NSO.1.5: match the powers first]; 5.7×10^6 — the blank holds a coefficient, not a whole term
9	Devon writes $(1.9 \times 10^2) + (6.6 \times 10^4) = 8.5 \times 10^6$. Name the two mistakes he made, and give the correct sum.	***6.619 × 10⁴***	'he added wrong' — no mistake is actually named [the item asks which two]; 8.5×10^4 — the exponent mistake was caught and the matching mistake was not

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the place-value strip from Notes I with the power written above each name, and let the student write both numbers as words first — 'four point nine five ten-thousands' and 'twelve point one ten-thousands' — before any symbol is written. The matching step becomes reading, not procedure.

ELL. Coefficient and power of ten are the two phrases every question uses. Write 7.7×10^4 once with COEFFICIENT under the 7.7 and POWER OF TEN under the 10^4, and point at the labels on every board. Add or subtract is the third phrase: read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4.2 \times 10^6) + (3.9 \times 10^5) + (1.1 \times 10^4)$ — three terms, all within two of each other — and then why the benchmark stops at a gap of two.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name both of Devon's mistakes separately and give 6.619×10^4. Read the boards, not the papers.

Vocabulary. **scientific notation**; **coefficient**; **common power of ten**

Materials. Whiteboards and markers. Calculators are allowed on everything; entering 8E−6 + 3E−6 and reading 1.1E−5 back off the display is worth thirty seconds at whiteboard 3.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Add and subtract numbers written in scientific notation — HUR.

Multiplying and Dividing in Scientific Notation

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 2–3 — Multiplying and Dividing in Scientific Notation	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.5

2. Standards

MA.8.NSO.1.5 Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

Must. Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

Must not. No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the product is regrouped before it is evaluated, so 'multiply the fronts, add the backs' is read off the structure rather than memorised as a rule.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one move each, written and up on the cue; dividing in scientific notation was the lowest single figure on the last state report.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2×10^{10} and 20×10^{9}, which are the same number, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Essential question. Why must the powers of ten match to add, but not to multiply?

Building on. 4.01 — the coefficient rule and the finishing step; MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.03) — the product and quotient of powers laws.

Working toward. 4.04 — real-world problems and significant digits; MA.912.NSO.1 — operations with rational exponents in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	12	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why do the powers of ten have to match before you add, but not before you multiply?	DOK 3
Marcus's power of ten is right and Sara's coefficient is right. Which of the two is closer to the correct answer, and does 'closer' mean anything useful here?	DOK 3
A quotient of two positive numbers comes out as 5.3×10^{-3}. What does the negative exponent tell you about how the two numbers compared?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$(3\times10^5)(3\times10^6)$	***9 × 10¹¹***	9 × 10³⁰ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6 × 10¹¹ — the coefficients were added [8.NSO.1.5: the operation is multiplication]
2	A printer makes 4 × 10³ copies per hour. It runs for 2.5 × 10² hours. Paper costs 3 cents a sheet. How many copies does it make?	***1 × 10⁶***	anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10 × 10⁵ — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception]
3	$\frac{9.6\times10^8}{1.2\times10^3}$	***8 × 10⁵***	8 × 10¹¹ — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4 × 10⁵ — the coefficients were subtracted instead of divided [8.NSO.1.5]
4	$\frac{3.28\times10^7}{4\times10^{10}}$	***8.2 × 10⁻⁴***	0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]; 8.2 × 10⁻³ — the point moved but the exponent did not
5	$(4 \times 10^7)(2 \times 10^9)$. Which is the product?	***C — 8 × 10¹⁶***	(A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication]
6	$(5 \times 10^6)(4 \times 10^3)$. Which is the product, written in scientific notation?	***D — 2 × 10¹⁰***	(A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12 × 10⁹ instead of 1.2 × 10¹⁰]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5]
7	$\frac{7\times10^9}{7\times10^{-6}}$	***1 × 10¹⁵***	1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1 × 10⁻¹⁵ — bottom minus top [8.NSO.1.5: top minus bottom]
8	$(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$. What is the value of n?	***n = 2***	4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2
9	For $(7.2 \times 10^{11}) \div (2.4 \times 10^3)$, Marcus writes 4.8 × 10⁸ and Sara writes 3 × 10¹⁴. Name what each one did wrong, and give the correct quotient.	***3 × 10⁸***	'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8 × 10⁸ — Marcus's error repeated

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-column organiser — COEFFICIENTS on the left, POWERS OF TEN on the right — and have the student split every expression into the two columns before computing anything. The regrouping step in Notes I becomes a physical act, and the two laws are then applied one column at a time.

ELL. Product and quotient are the two nouns, multiply and divide the two verbs. Write them in pairs on the board — PRODUCT goes with MULTIPLY and ADD the exponents, QUOTIENT goes with DIVIDE and SUBTRACT the exponents — and read each board's operation aloud before the boards go up.

Enrichment. Ask for $(4 \times 10^{5})^{2}$ without a calculator (1.6×10^{11}), then for the two numbers in scientific notation whose product is 1.2×10^{7} and whose quotient is 1.2×10^{1} (1.2×10^{4} and 1×10^{3}).

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name Marcus's error and Sara's error separately and give 3×10^{8}. Read the boards, not the papers.

Vocabulary. **product of powers law**; **quotient of powers law**; **regrouping a product**

Materials. Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether 8×10^{16} and 8×10^{63} could possibly both be the answer to whiteboard 5.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Multiply numbers written in scientific notation — YZU; Divide numbers written in scientific notation — SGT.

Real-World Problems and Significant Digits

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 4 — Real-World Problems and Significant Digits	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.6

2. Standards

MA.8.NSO.1.6 Solve real-world problems involving operations with numbers expressed in scientific notation.

Must. Solve real-world problems with numbers in scientific notation, and recognise the importance of significant digits when physical measurements are involved.

Must not. The within-2 exponent limit still applies to addition and subtraction here. Significant digits limit a reported answer; they never change the arithmetic, and a count is never a measurement.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 2, 6 and 7 — the last move on every board is asking how much of the answer the measurements actually support, before the board goes up.
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Every problem from Notes II on — each number is a measured quantity with a unit, because the rule has no meaning without one.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits between 2.6×10^5 and 2.61×10^5, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Essential question. How much of an answer is worth writing down, when the numbers that produced it were measured?

Building on. 4.01–4.03 — all four operations in scientific notation; Grade 5 and 6 — place value and rounding to a named place.

Working toward. 4.05–4.08 — the order of operations with exponents and radicals; MA.912.GR.4 — measurement and precision in geometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why does the last digit of a measurement carry less information than the digits before it?	DOK 2
A sum of two two-digit measurements can have three significant digits. How is that possible, when a product of the same two could only have two?	DOK 3
Ellis uses the product rule on a sum and throws away two digits that were measured. What is each of the two rules actually protecting?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	How many significant digits are in 0.00470 ?	***3***	2 — the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]; 5 — the leading zeros were counted [B1G-M 8.NSO.1.6: leading zeros are never significant]
2	A pond's surface measures 1.4×10^2 m by 8.5×10^1 m. The pond is 3 meters deep. Find the surface area. Round to the correct number of significant digits.	***1.2×10^4 square meters***	anything built from the depth — the unneeded figure was used [an area needs two dimensions]; 1.19×10^4 — the arithmetic without the rounding [8.NSO.1.6 clarification]
3	Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5. How should the answer be written?	***7.41×10^5***	7.4×10^5 — two digits, the wrong one of the two counts [8.NSO.1.6: the answer takes the smaller count, which is 3]; 7.4126×10^5 — no rounding at all
4	$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured. Write the sum with the correct precision.	***3.0×10^3 m***	2.971×10^3 — the arithmetic without the precision rule [8.NSO.1.6 clarification]; 2.9×10^3 — 2,971 rounded down instead of up [rounding to the hundreds place]
5	How many significant digits are in 0.03080?	***A — 4***	(B) only the nonzero digits were counted [B1G-M 8.NSO.1.6: a zero between nonzero digits is always significant]; (C) one of the leading zeros was counted [B1G-M 8.NSO.1.6: leading zeros are never significant]; (D) the trailing zero was dropped [B1G-M 8.NSO.1.6: a trailing zero after a decimal point is significant]
6	A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm. Which is the area, to the correct significant digits?	***B — 2.6×10^5 cm^2***	(A) the exact product, not reduced, and 26.1 is not less than 10 — two mistakes [B1G-M 8.NSO.1.5 misconception: leaving 12×10^9 instead of 1.2×10^{10}]; (C) the exponent was not raised when the coefficient was repaired [8.NSO.1.5: point one place left, exponent up one]; (D) the exact product, not reduced to the two significant digits the measurements carry [8.NSO.1.6 clarification]
7	A car travels 4.20×10^2 km in 3.0 hours. What is its average speed, to the right number of significant digits?	***1.4×10^2 km/h***	140.0 — four digits claimed from a two-digit measurement [8.NSO.1.6 clarification]; 1.4×10^1 — the exponent is one too low
8	A measured length times a measured width gives an area of 3.6×10^3 m^2. The length measured 9.0×10^1 m. What did the width measure, to two significant digits?	***4.0×10^1 m***	4×10^1 — one significant digit where two were asked for [8.NSO.1.6: a trailing zero after a decimal point carries the second digit]; 4.0×10^2 — the exponent is one too high
9	Ellis says that because 9,145 has 4 significant digits and 4.8 has 2, the sum 9,145 + 4.8 must be written with 2 significant digits, as 9.1×10^3. Explain why he is wrong, and give the correct sum.	***9,150 = 9.150×10^3***	'he used the wrong rule' — the right rule is not stated [the item asks for an explanation]; 9,149.8 — no rounding at all, which claims a tenth that 9,145 never measured

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four counting rules as a card with one worked number each, in the order of Notes I, and let it stay on the desk through the round. On every problem the student marks the significant digits on the two measurements before doing any arithmetic, so the count is decided before it is needed.

ELL. Significant, precise and measurement are three words doing specific work today. Put MEASUREMENT = a number someone measured beside COUNT = a number someone counted, with 4.20 meters under one and 24 students under the other, and ask which kind each number is before every problem.

Enrichment. Ask what 1,000 means when it is written with no decimal point — one significant digit or four — and how a scientist would write each of the two. Then ask why the state guide's own Puerto Rico item is genuinely ambiguous.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say what the rule for a sum actually is, not only that Ellis used the wrong one, and give 9.150×10^3. Read the boards, not the papers.

Vocabulary. **significant digits**; **leading zero**; **measurement**

Materials. Whiteboards and markers. Calculators are allowed on everything and are genuinely useful today, because the point of the lesson is what to do with a long display rather than how to produce one.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S.

Evaluating Expressions with Radicals

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 5 — Evaluating Expressions with Radicals	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate expressions containing square roots, cube roots and integer exponents, simplifying radicals by factoring within the benchmark's lists.

Must not. Perfect squares up to 225 and perfect cubes from −125 to 125, or fractions whose parts are each in those lists. Estimating a root that is not perfect was Unit 2. No fractional exponents.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the radical bar is recognised as a grouping symbol by comparing √(9 + 16) with √9 + √16, so the rule is derived from the difference rather than announced.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Boards 3 and 6 — students say whether a negative answer is possible at all before they evaluate, which is the whole square-root-versus-cube-root distinction.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 13 and 17 on √(25 + 144), the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate an expression that contains square roots, cube roots and integer exponents.

Essential question. What does the radical bar do to the order in which an expression is evaluated?

Building on. Unit 2 (2.01–2.06) — perfect squares, perfect cubes and the principal square root; 3.04 — negative exponents; Grade 6 — the order of operations.

Working toward. 4.06 — the whole order of operations at once; 4.07–08 — real-world multi-step problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is $\sqrt{(9 + 16)}$ not the same as $\sqrt{9} + \sqrt{16}$, when $(9 + 16) \div 5$ is the same as $9 \div 5 + 16 \div 5$?	DOK 3
A cube root of a negative number is ordinary and a square root of one is not. What is it about the exponent 2 that makes the difference?	DOK 3
$\sqrt{49}$ is 7, but the equation $x^2 = 49$ has two solutions. Are those two statements in conflict?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$\sqrt{121}$	***11***	60.5 — the radicand was halved [Unit 2, 2.02: a root is not a half]; ±11 — the radical sign names the principal root only [8.NSO.1.7]
2	A square patio has an area of 169 square feet. Fencing for it costs 12 dollars a foot. How long is one side of the patio?	***13 feet***	anything built from 12 dollars — the unneeded figure was used [the question asks for a length, not a cost]; 52 — the perimeter, not a side; 84.5 — the area was halved
3	$\sqrt[3]{-125}$	***−5***	5 — the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]; 'not a real number' — the square-root rule applied to a cube root [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]
4	$\sqrt{2^3 + 17}$	***5***	25 — the arithmetic under the bar was done and the root was not [8.NSO.1.7]; anything with two separate roots — the bar groups, so it is one radicand [8.NSO.1.7]
5	√(25 + 144). Which is the value?	***D — 13***	(A) the root of each piece was taken and the results added, 5 + 12 [8.NSO.1.7: the bar groups, so the radicand is 169]; (B) the sum was found and the root was never taken [8.NSO.1.7]; (C) the radicand was halved instead of rooted [Unit 2, 2.02: a root is not a half]
6	The cube root of −1/64. Which is the value?	***A — −1/4***	(B) the square-root rule for negatives was applied to a cube root [B1G-M 8.NSO.1.7 misconception]; (C) the square root of 1/64 was taken and the sign kept [B1G-M 8.NSO.1.7 misconception: confusing square roots with cube roots]; (D) the sign was dropped [8.NSO.1.7: an odd root keeps the sign of its radicand]
7	$\sqrt{\frac{4}{9}} + \sqrt[3]{\frac{8}{27}}$	***4/3***	12/36 — the roots were never taken, then 4/9 + 8/27 was added top-and-bottom [two errors: the roots, then Grade 5 fraction addition]; 2/3 — only one of the two radicals was evaluated
8	$\sqrt{5^3 - 25} \cdot 10^{-3}$	***1/100***	10,000 — 10^{-3} read as 10^3 [8.NSO.1.3: a negative exponent is a reciprocal]; −1/100 — the exponent's sign made the value's sign [B1G-M 8.NSO.1.3 task: −b vs b^{-1}]
9	Simon says that the cube root of −27 is not a real number, because you cannot take the root of a negative. Explain why he is wrong, and give the value.	***−3***	'−3' alone — the value without the reason [the item asks why he is wrong]; '3' — the sign dropped as well as the rule misread

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the perfect-square and perfect-cube lists from Unit 2's reference sheet and let them stay out; a root is a lookup today, not a recall. On every board with arithmetic under the bar, the student draws a box around the radicand before writing anything, so the grouping is a physical act.

ELL. Radicand and radical are the two new nouns, and they are easy to confuse because they sound alike. Write √25 once with RADICAL circling the whole symbol and RADICAND labelling only the 25, and point at the labels on every board.

Enrichment. Ask for a pair of numbers for which √(a + b) really does equal √a + √b, and what has to be true of them. Then ask why the same question for a product has a different answer.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to say which root Simon's rule belongs to, not only that he is wrong, and give −3. Read the boards, not the papers.

Vocabulary. **radical**; **radicand**; **principal square root**

Materials. Whiteboards and markers, and Unit 2's reference sheet if students still want the perfect-square list. Calculators are allowed on everything; a calculator will refuse the cube root of a negative on some models, which is worth showing once and explaining rather than trusting.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, this skill continues tomorrow: one assignment, due at the start of the class after 4.06: Evaluate radical expressions — D8J.

Order of Operations with Radicals

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lesson 6 — Order of Operations with Radicals	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Evaluate multi-step expressions with rational numbers, exponents and radicals, in six steps or fewer.

Must not. No mnemonic. The state guide says to avoid PEMDAS because it leaves out grouping symbols, and the word appears on no student page in this unit.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — the four levels are read off the expression's structure and the class is never given a mnemonic, because the guide asks us not to give one.
MA.K12.MTR.3.1	Complete tasks with mathematical fluency.	Boards 1, 3 and 7 — one structural decision each, written and up on the cue; these are the errors that cost marks everywhere else.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 5 — when the room splits between 50 and 80, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Essential question. What decides the order in which an expression is evaluated?

Building on. 4.05 — the radical bar as a grouping symbol; 3.04–3.05 — negative exponents on integer and rational bases; Grade 6 — the order of operations with whole numbers.

Working toward. 4.07–08 — the same structure inside real-world problems; MA.912.NSO.1.4 in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is -2^2 different from $(-2)^2$? What is the exponent attached to in each one?	DOK 3
A mnemonic for the order of operations has no letter for the radical bar or the fraction bar. What does reading the structure give you that the mnemonic cannot?	DOK 3
Roscoe says a square is always positive. What would have to be true about $\sqrt{n} - 11$ for that to be right, and what value of n breaks it?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$90 \div 5 \cdot 3$	***54***	6 — $5 \cdot 3$ was done first [8.NSO.1.7: multiplication and division are one level, taken left to right]; 1,350 — the division was read as a multiplication
2	A square tile has side √196 centimeters. Its diagonal is about 19.8 centimeters. What is the perimeter of the tile?	***56 centimeters***	anything built from 19.8 — the unneeded figure was used [a perimeter needs the side]; 14 — the side, not the perimeter; 196 — the root was never taken
3	$16 - 5 + 4$	***15***	7 — $5 + 4$ was done first [8.NSO.1.7: addition and subtraction are one level, taken left to right]
4	$\frac{3^3 - 21}{5^2 - 15}$	***3/5***	anything that divides before the top and bottom are finished [8.NSO.1.7: the fraction bar groups both]; −6/5 — 5^2 read as $5 \cdot 2$, leaving 6 over −5; 6/25 — the −15 under the bar was dropped
5	$2 + 3 \cdot 4^2$. Which is the value?	***B — 50***	(A) the exponent was applied to $3 \cdot 4$ instead of to the 4 alone [8.NSO.1.7: an exponent belongs to its own base]; (C) 4^2 was read as $4 \cdot 2$ [8.NSO.1.3: an exponent is not a factor]; (D) $2 + 3$ was done first, then $5 \cdot 16$ [8.NSO.1.7: multiplication comes before addition]
6	$(6^2 - 4)$ over 2^3. Which is the value?	***A — 4***	(B) only the 4 was divided by 2^3 before the subtraction [8.NSO.1.7: the bar groups the whole numerator]; (C) 6^2 was divided by 2^3 first and then 4 was subtracted [8.NSO.1.7: the bar groups the whole numerator]; (D) the numerator was multiplied by 2^3 instead of divided [8.NSO.1.7: the bar is division]
7	$-2^2 + (-2)^2$	***0***	8 — both terms read as $(-2)^2$ [8.NSO.1.7: an exponent attaches to its own base, and parentheses decide what that base is]; −8 — both terms read as -2^2
8	$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$	***17/6***	12/8 + 12/9 left unadded — a value, not an answer; 1/8 + 1/9 = 2/17 — the denominators were added [Grade 5: adding fractions needs a common denominator]
9	Roscoe says that $(\sqrt{n} - 11)^2$ is always positive, because a square is always positive. Is he right? Give a value of n that settles it.	***No — at n = 121 the value is 0***	'yes, squares are positive' — the claim repeated rather than tested [8.NSO.1.7: a square is never negative, which is a different statement]; 'no' with no value of n — the counterexample is the evidence

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the four-level table from Notes I as a card and let it stay out. On every expression the student first circles all the grouping symbols and numbers them from the inside out, and only then evaluates; the circling is the work, and the arithmetic follows it.

ELL. Grouping symbol is the phrase to own. Put the four of them on the board — parentheses, brackets, the radical bar, the fraction bar — and have the student point to each one in an expression and name it before evaluating anything.

Enrichment. Ask for an expression using 2, 3 and 4 once each, with any grouping symbols, whose value is as large as possible — then as small as possible. Then ask why a mnemonic could not have answered either question.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give a value of n that settles Roscoe's claim and say what is wrong with the sentence, not only that it is wrong. Read the boards, not the papers.

Vocabulary. **grouping symbol**; **nested**; **multi-step expression**

Materials. Whiteboards and markers. Calculators are allowed on everything, and this is the lesson where a calculator is worth using against the class: enter -2^2 and $(-2)^2$ and let the machine settle it.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate radical expressions — D8J.

Real-World Order of Operations

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	4 — Solving Problems with Rational Numbers
Lesson	Lessons 7–8 — Real-World Order of Operations	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.NSO.1.7

2. Standards

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must. Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not. No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 — the expression arrives inside a situation, and the unit is part of the answer.
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II and board 9 — the same three numbers are written as two different expressions, so the structure is seen to come from the situation and not from the page.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits over whether the flat fee goes inside the parentheses, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Essential question. If key words cannot tell you which operation a problem needs, what can?

Building on. 4.05–4.06 — grouping symbols and the four levels; MA.6.AR.1 — substituting a value for a letter; Unit 2 — rounding at the end, not in the middle.

Working toward. The Unit 4 assessment; MA.8.AR.2 and MA.912.NSO.1.4, where formulas are rearranged rather than only evaluated.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	12	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Two problems have the same numbers and the same words and different answers. What is it in the problem, then, that decides the operation?	DOK 3
Why does rounding √200 to 14 before multiplying by 6 lose most of a foot, when 14 is only about 0.14 away from √200?	DOK 3
In a cost expression, what has to be true of a price for it to belong inside the parentheses that are multiplied by the area?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	Evaluate $3x^2$ when $x = -2$.	***12***	−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]
2	A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is K = ½mv², for a mass m and a speed v. Find her kinetic energy.	***2,240 joules***	anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules
3	Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours.	***44 dollars***	78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]
4	A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost.	***284 dollars***	4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area
5	Evaluate ½mv² when m = 40 and v = 3. Which is the value?	***A — 180***	(B) v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, $(½ \cdot 40 \cdot 3)^2$ [8.NSO.1.7: the exponent belongs to the v alone]
6	A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost?	***D — $12^2(1.25) + 38$***	(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate]
7	A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot.	***48.99 feet***	48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet
8	Evaluate $(a^2 - b) \div c$ when a = 7, b = 4, c = 5.	***9***	48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted
9	One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you.	***23 dollars and 31 dollars***	23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give a two-step frame — WRITE THE EXPRESSION, then EVALUATE IT — and require the expression on paper before any arithmetic. Substituted values go in parentheses every time, with the parentheses pre-printed on the frame for the first two problems.

ELL. Rate and flat fee are the two phrases every cost problem turns on, and 'a square foot' is what marks the difference. Put 'dollars a square foot' and 'a flat 60 dollars' side by side on the board and ask which one depends on the size of the floor, before any expression is written.

Enrichment. Ask for a third problem using 15, 4 and 6 whose answer is neither 39 nor 66. Then ask whether every arrangement of three numbers and two operations can be made into a sensible situation.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to give both totals and say what decides the operation, since the words and the numbers are identical in the two problems. Read the boards, not the papers.

Vocabulary. **formula**; **substitute**; **unit**

Materials. Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Evaluate variable expressions: word problems — DPG.